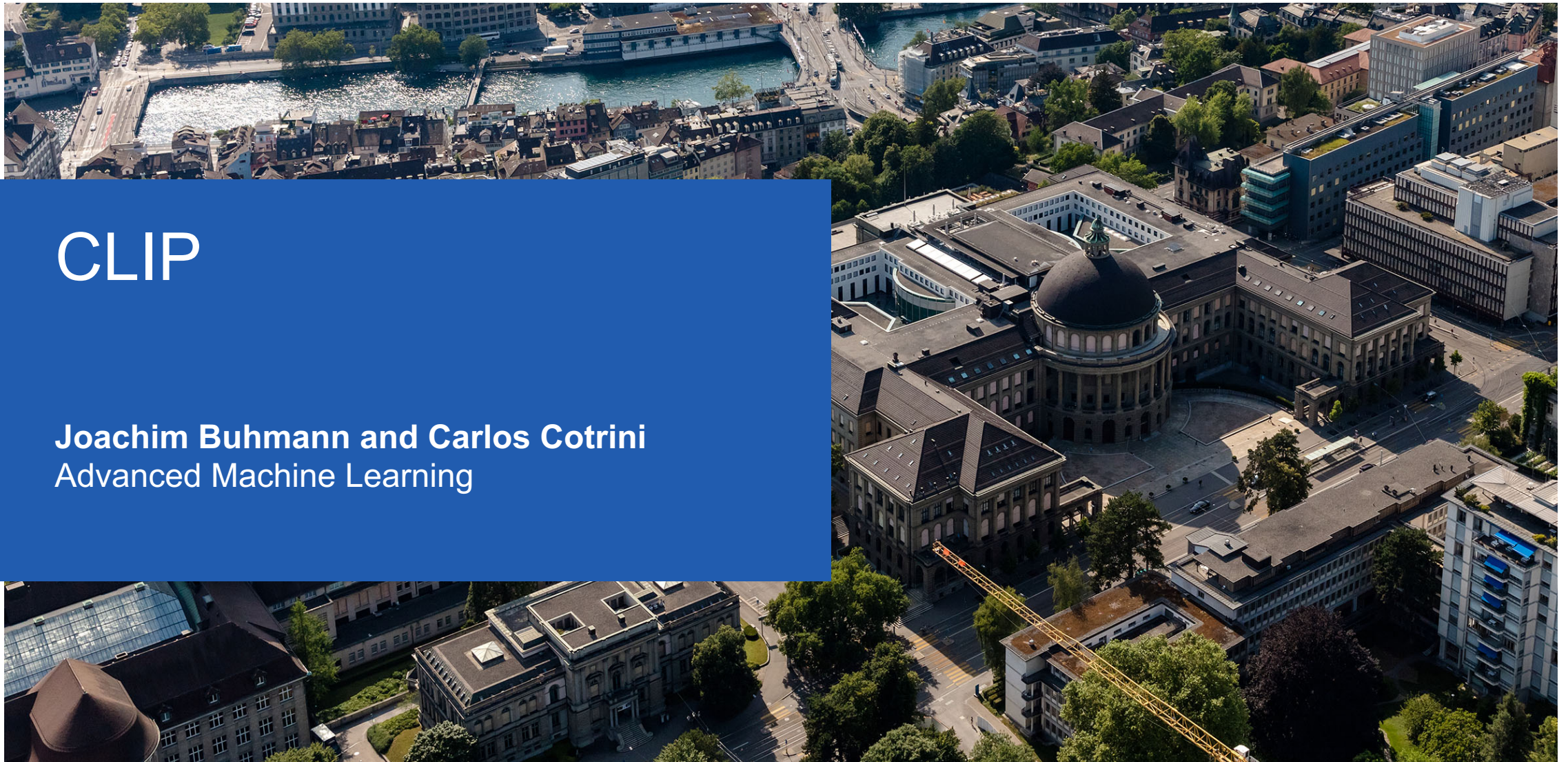
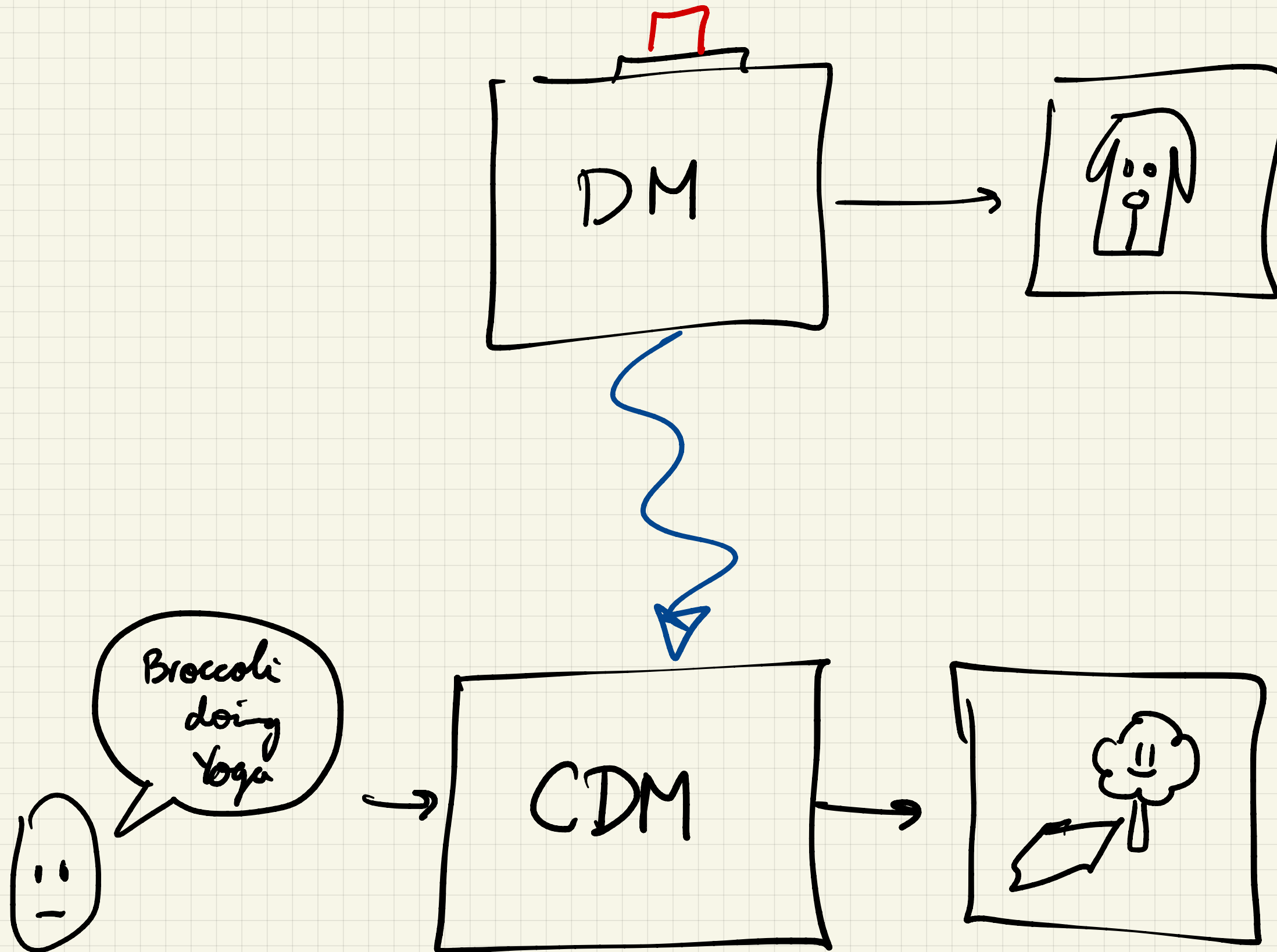
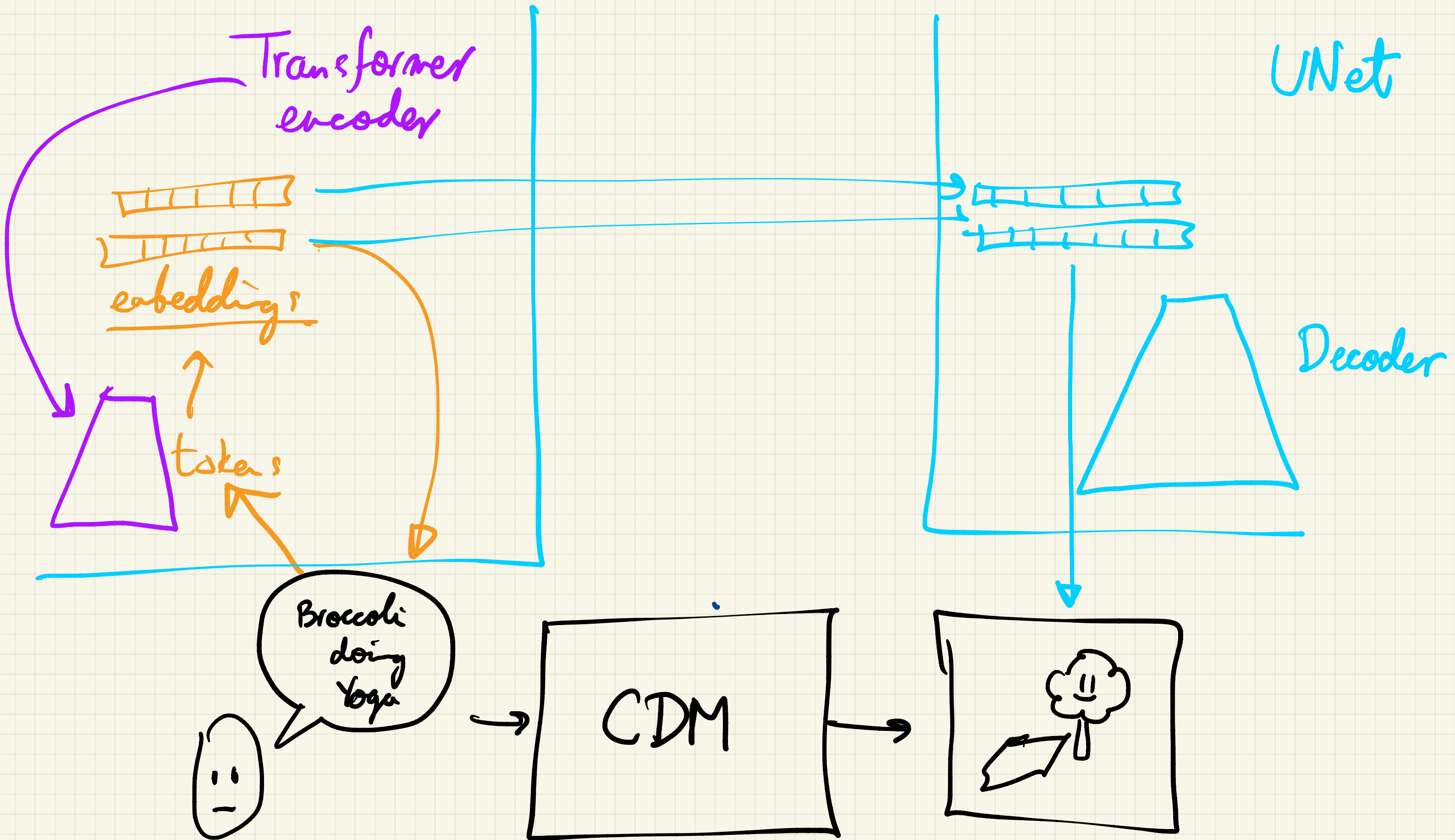


CLIP

Joachim Buhmann and Carlos Cotrini
Advanced Machine Learning

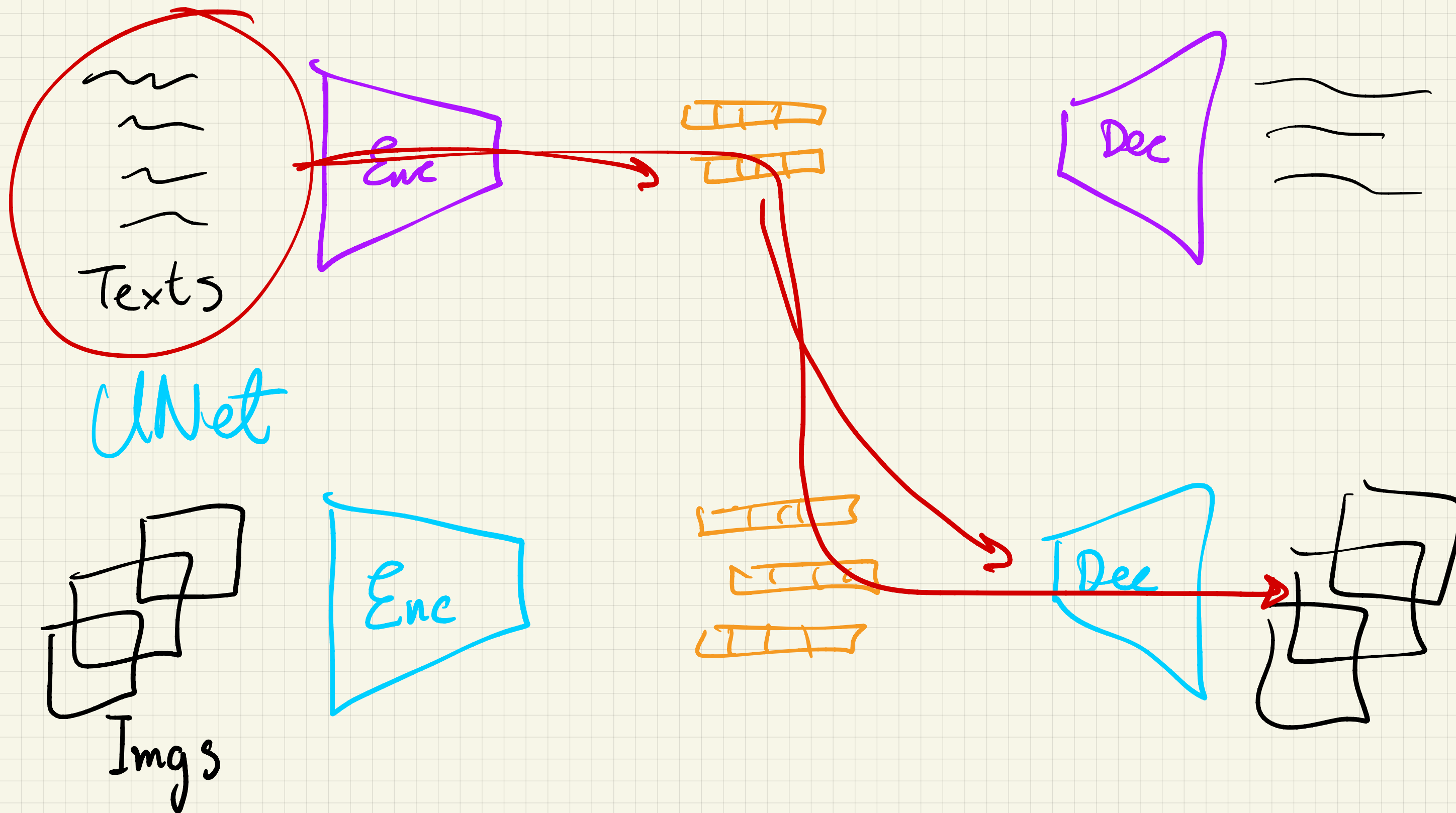






Transformer

Naive
sol.



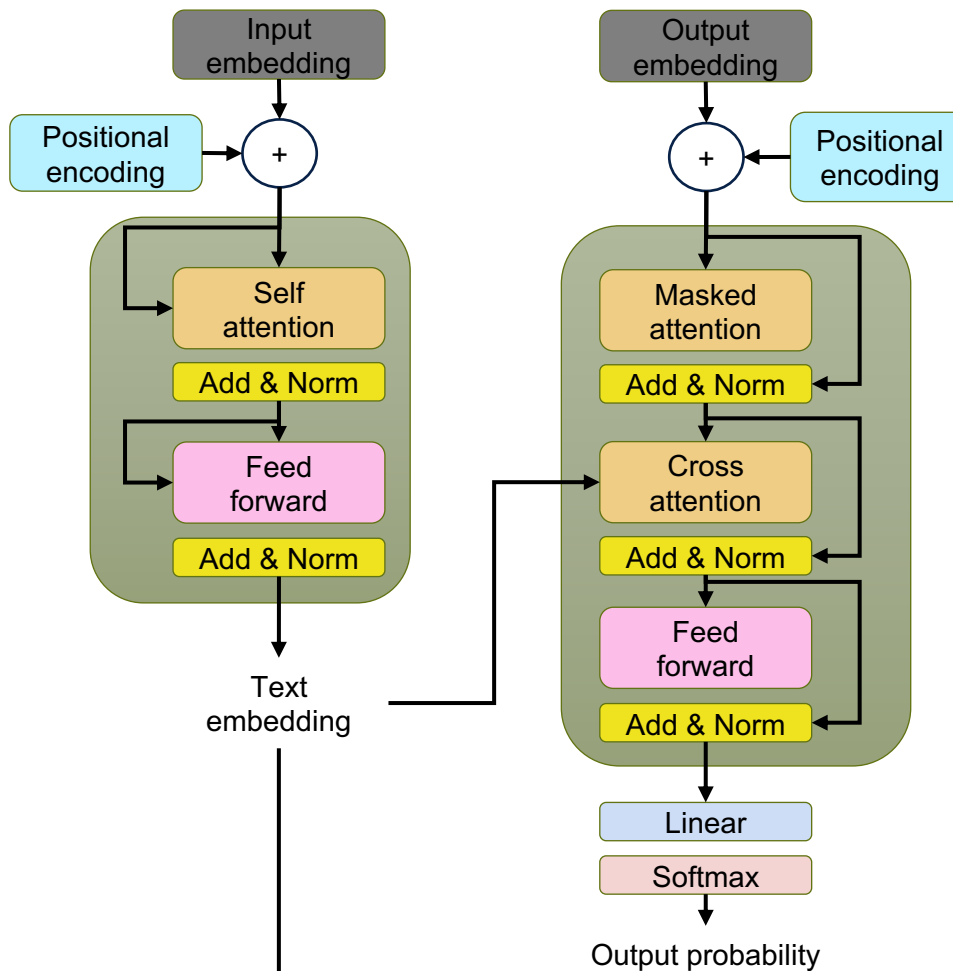
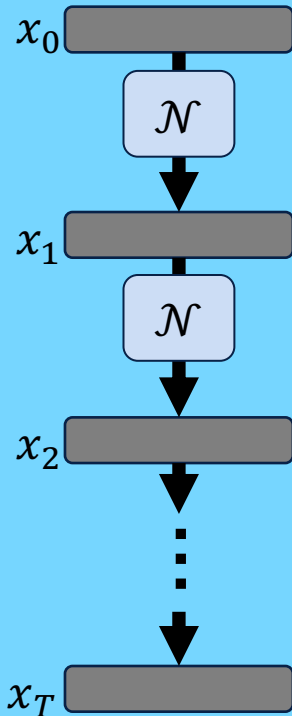


“A broccoli
doing yoga”

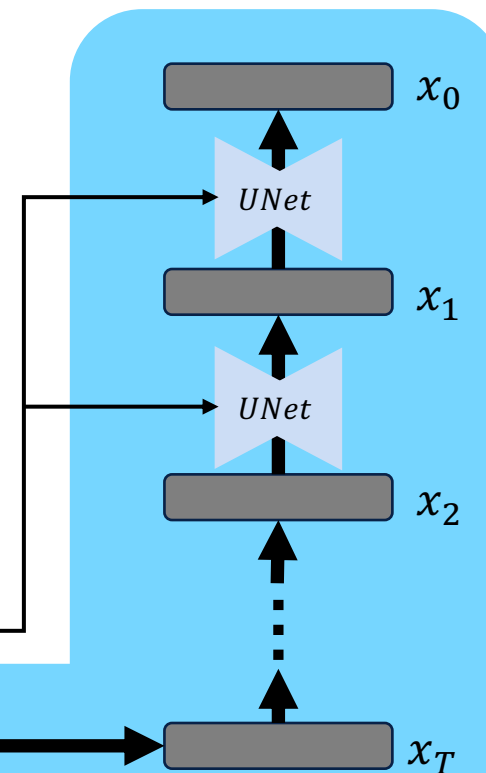
“Ein Brokkoli,
der Yoga macht”



VAE encoder



VAE decoder



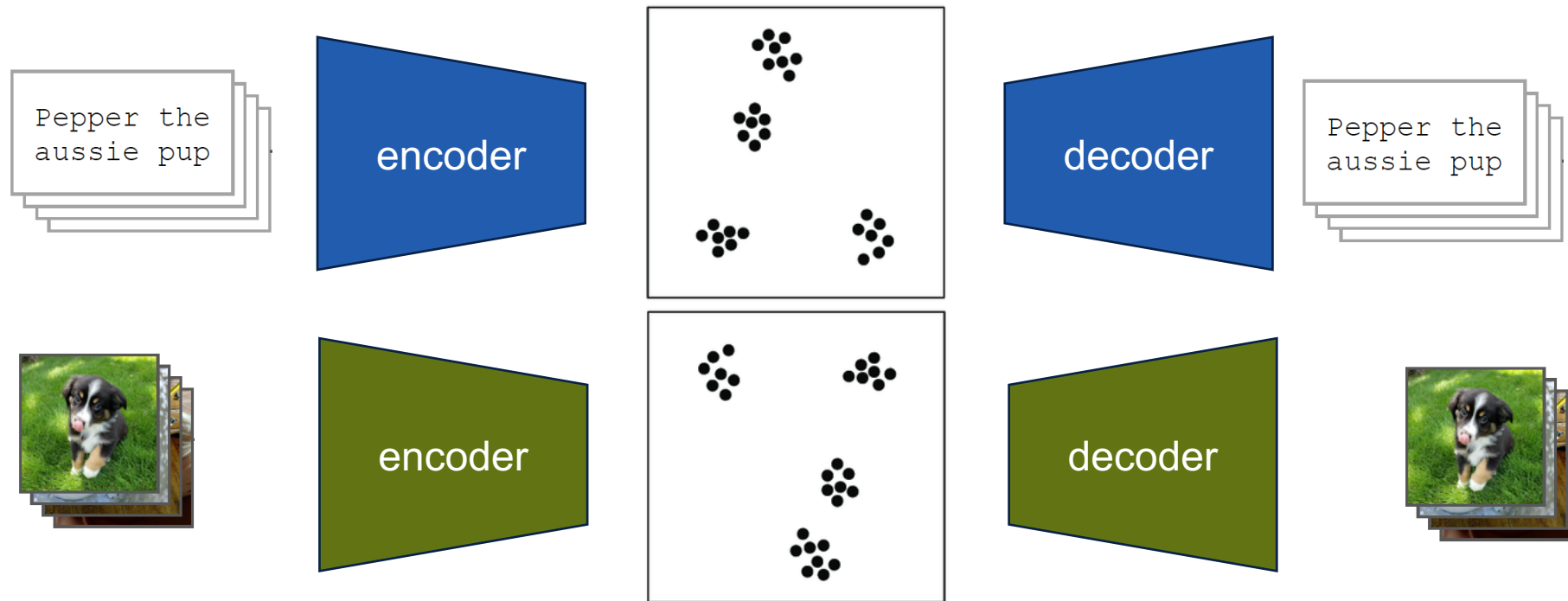
CLIP

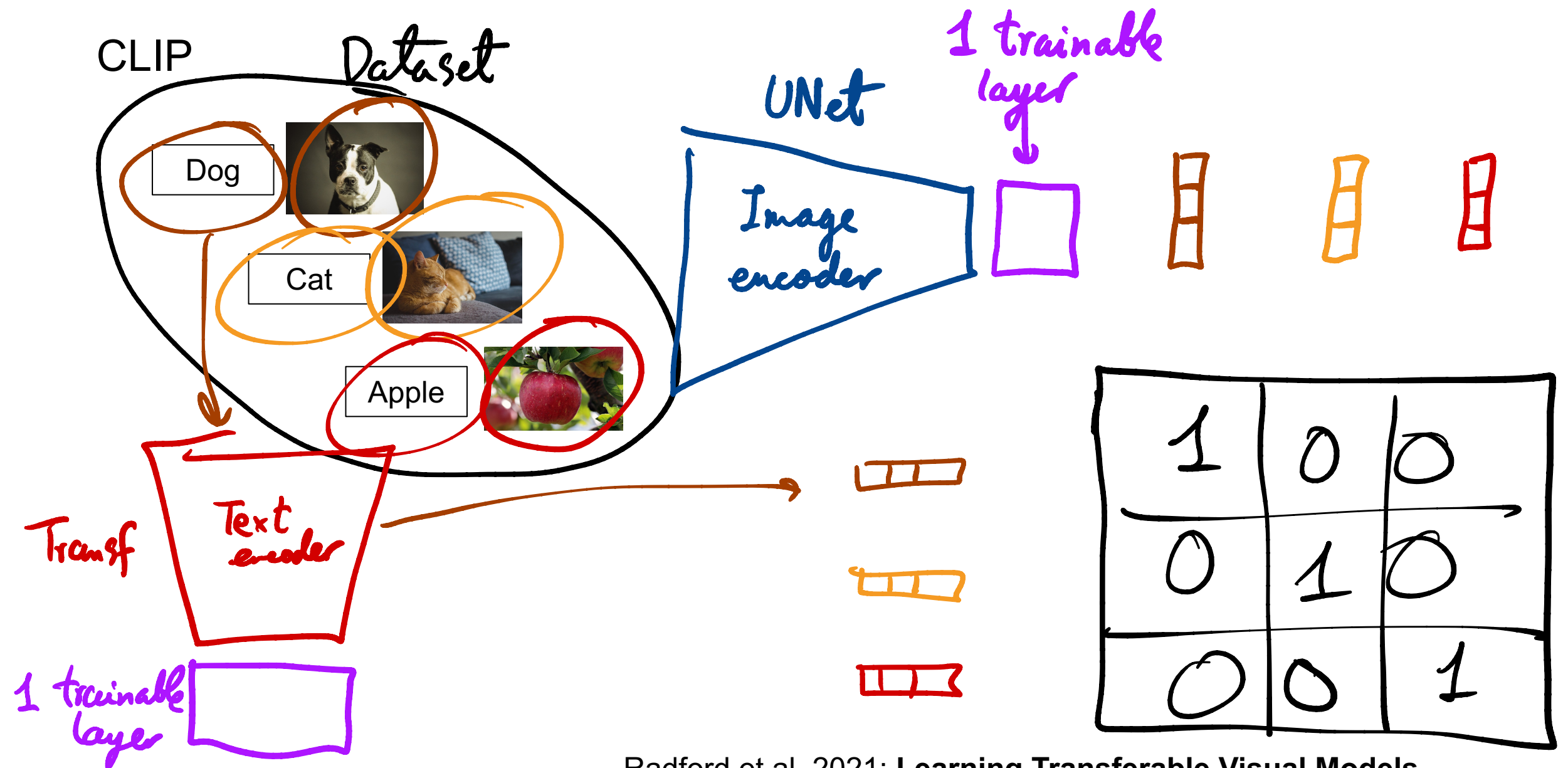
Autoencoders create their own representations

Representations produced by one autoencoder are seldom understandable for another autoencoder.

This prevents the decoding as text of image representations produced by an encoder.

How can we harmonize such representations?

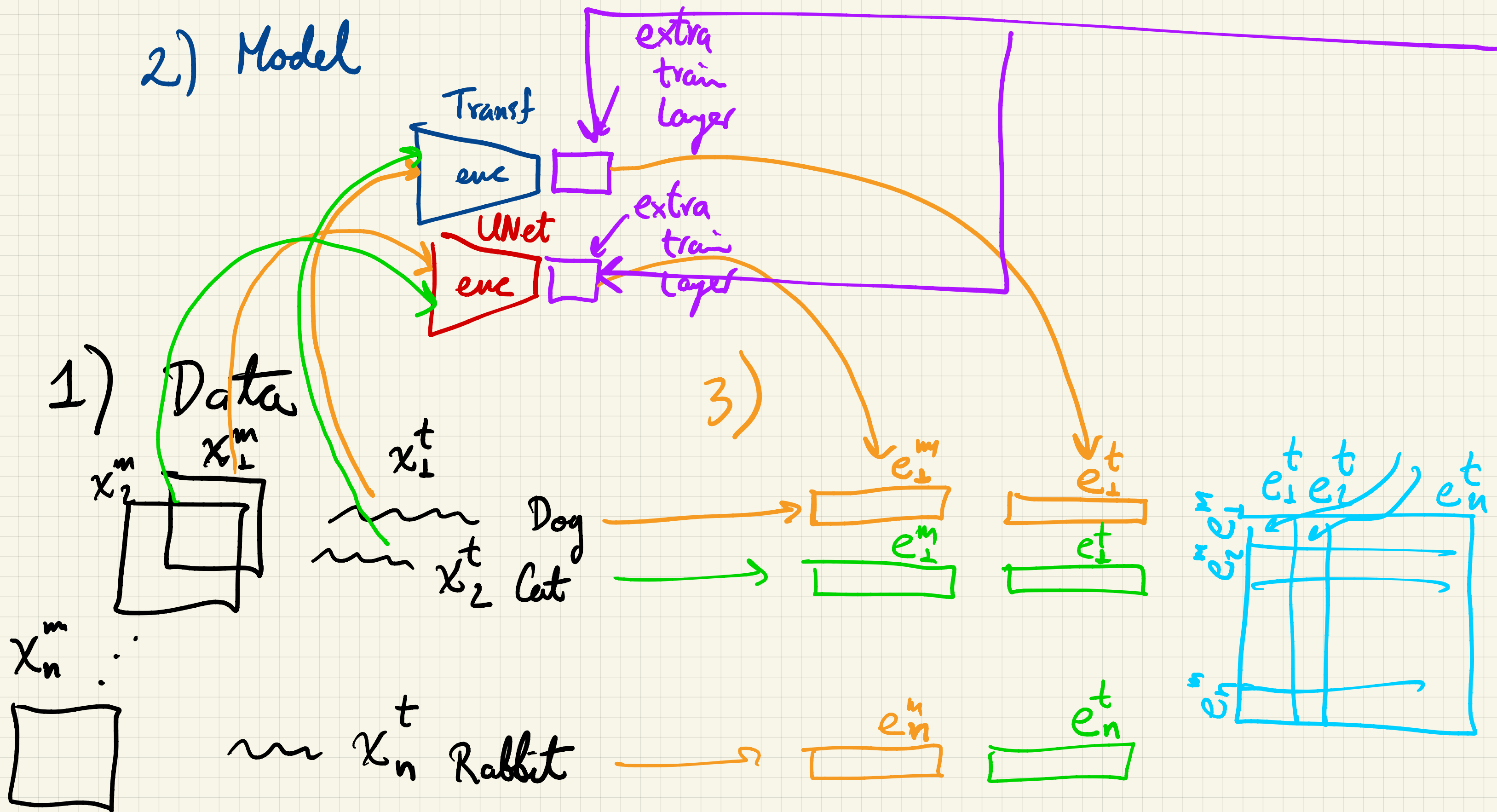




Radford et al, 2021: Learning Transferable Visual Models From Natural Language Supervision.

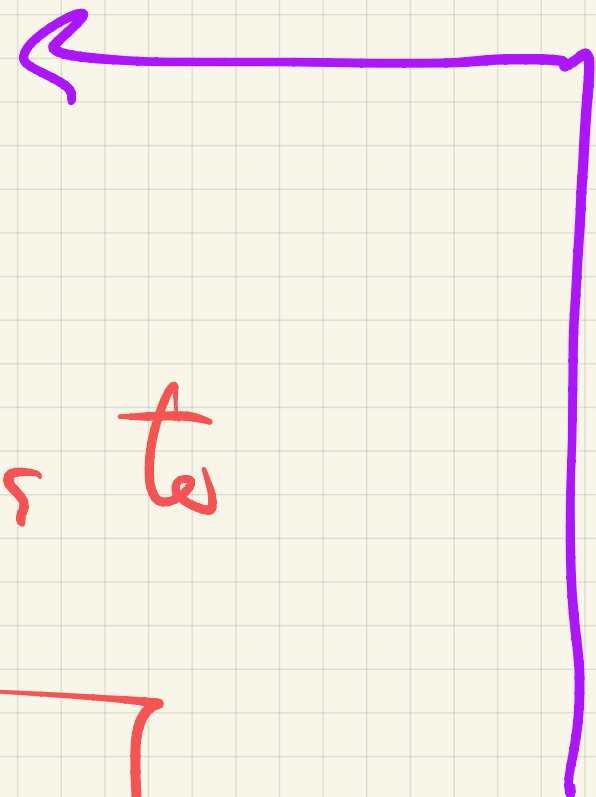
How to train CLIP?

2) Model



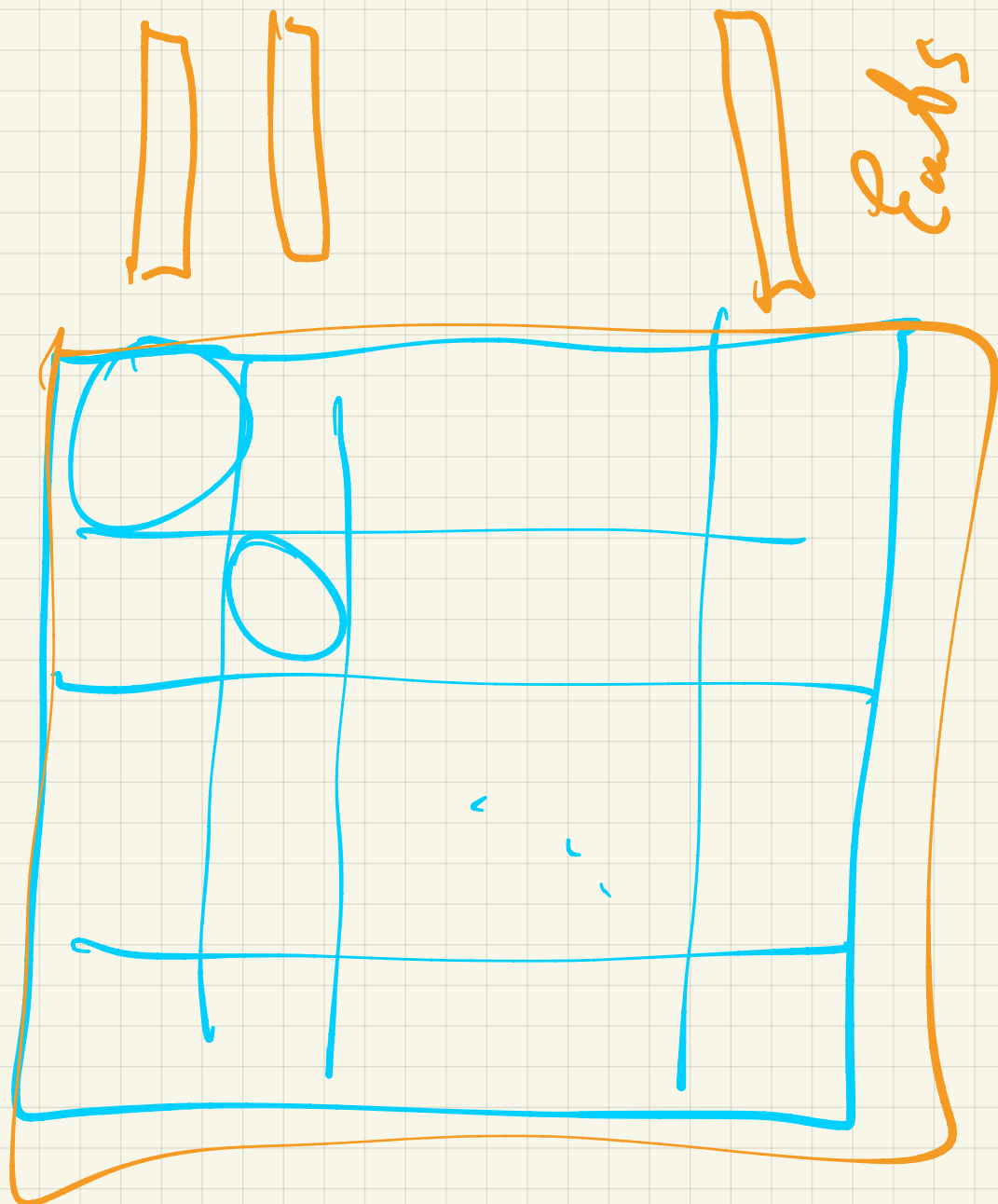
Computing the loss

$$\nabla \bar{\mathcal{L}}$$



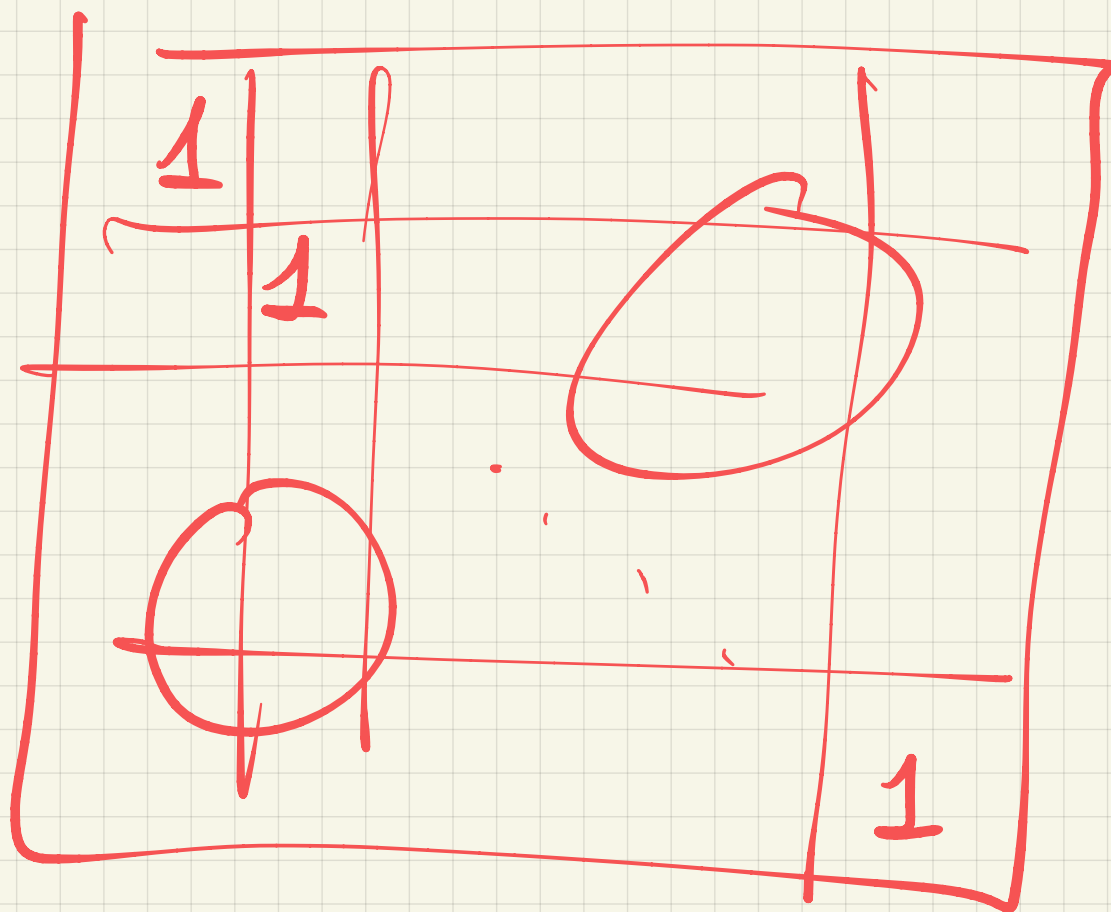
$$\bar{\mathcal{L}}$$

Ends
of
imgs



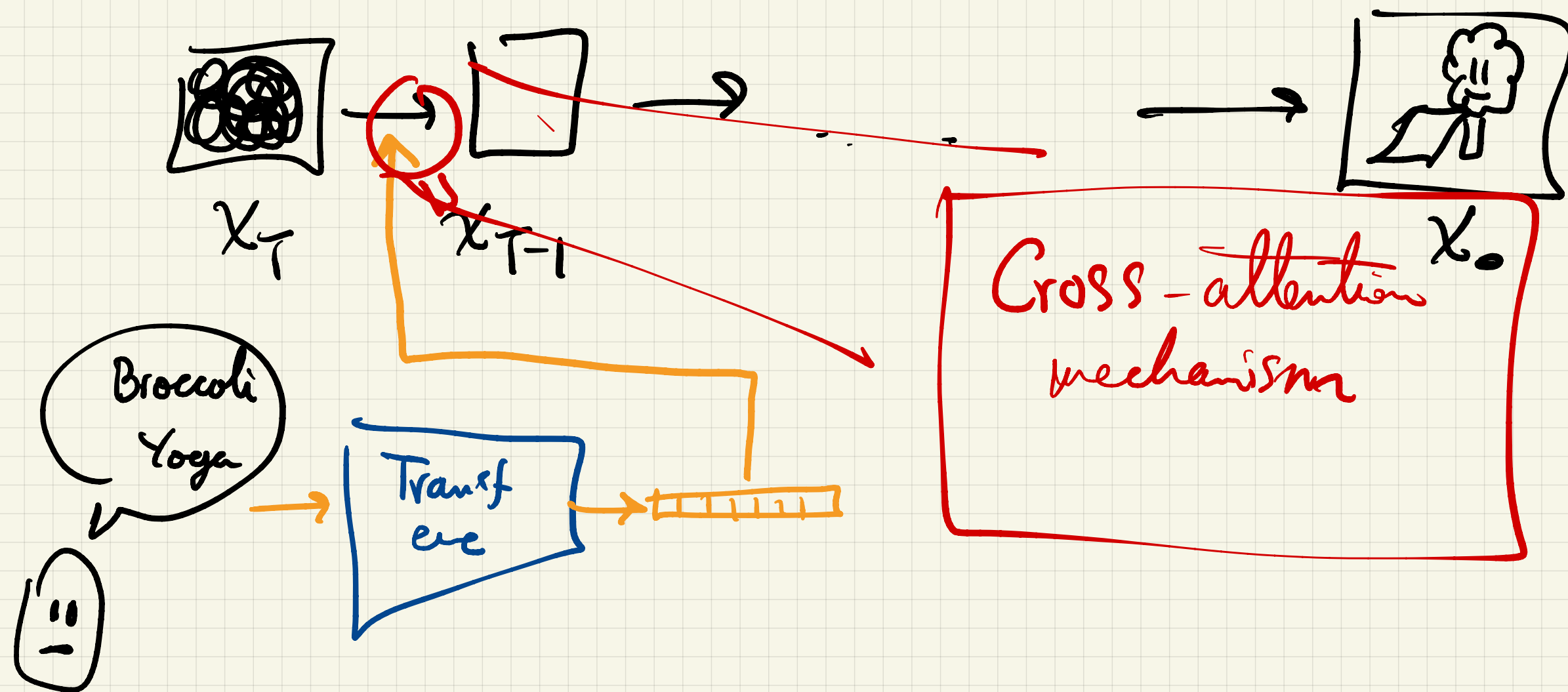
Ends of
text

How close this is to

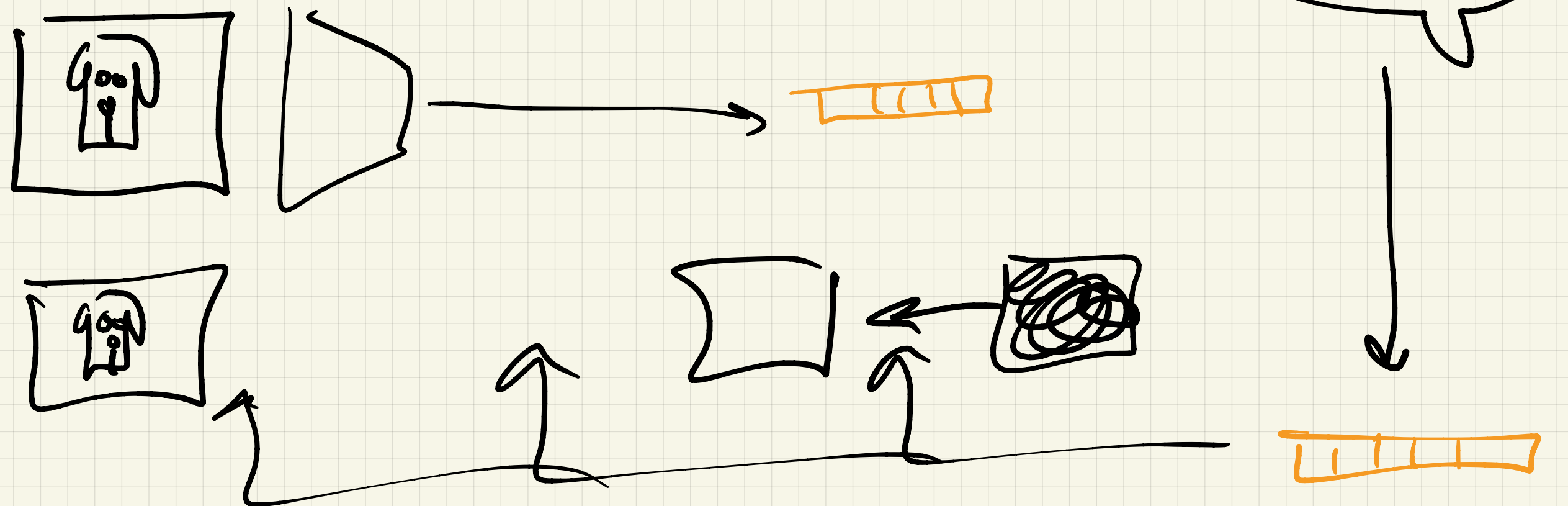


After the CLIP, the decoder or the image generator needs to be retrained.

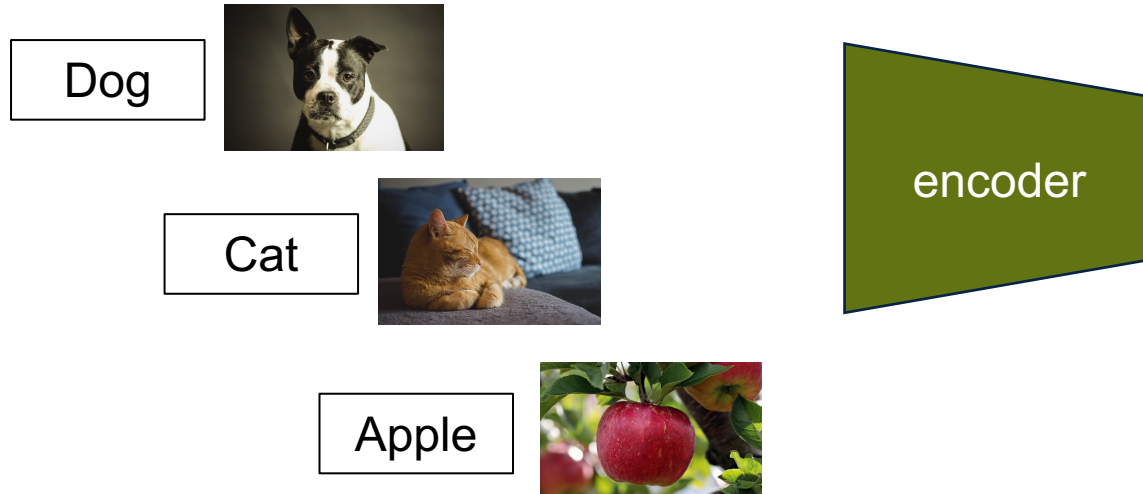
Example w/ diffusion model



Retrain ing you

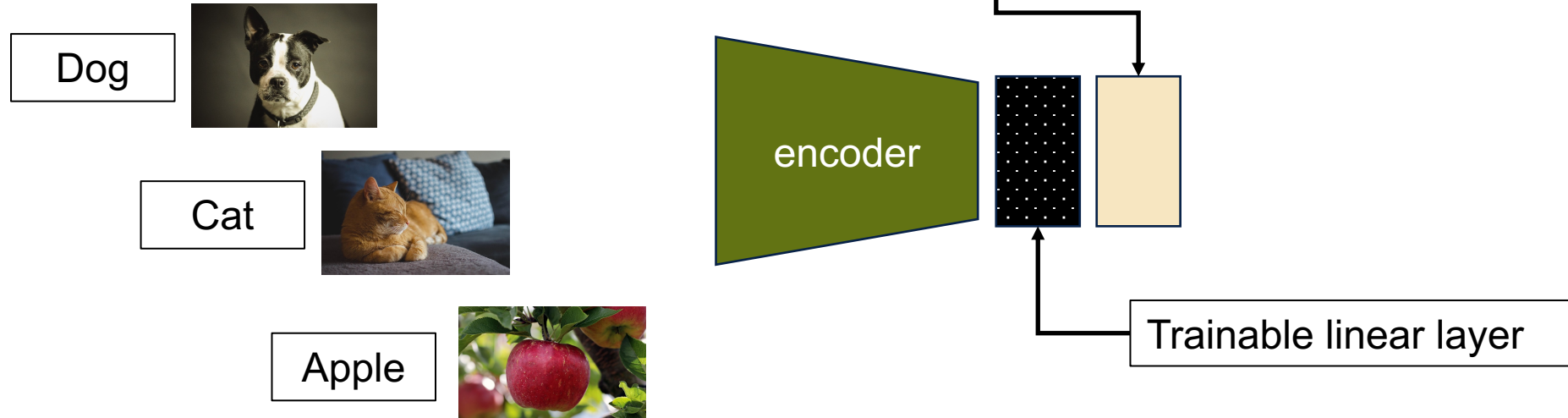


CLIP



Radford et al, 2021: **Learning Transferable Visual Models From Natural Language Supervision.**

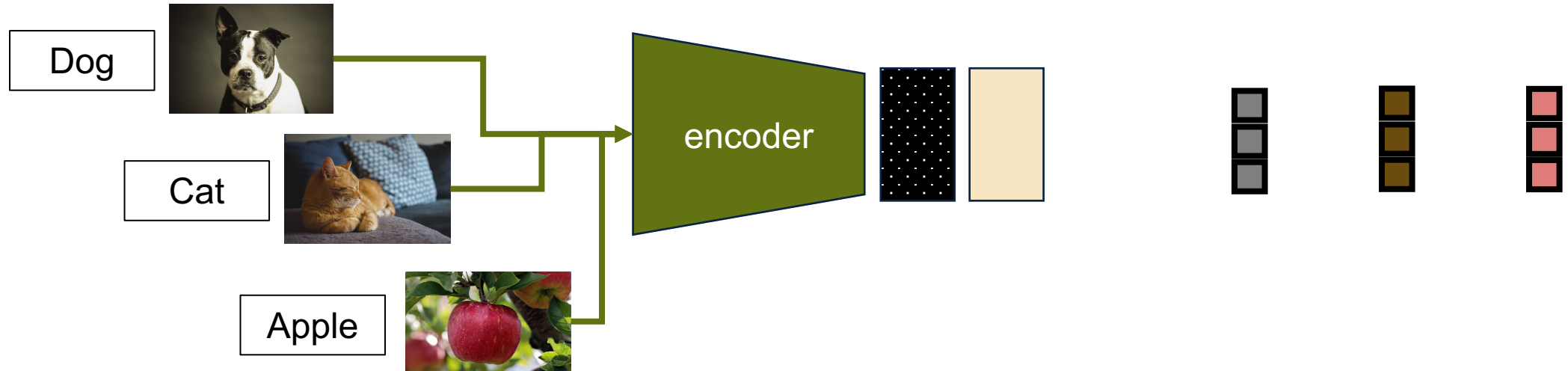
CLIP



Add a trainable linear layer at the end of the encoder and a normalization layer.

Radford et al, 2021: **Learning Transferable Visual Models From Natural Language Supervision.**

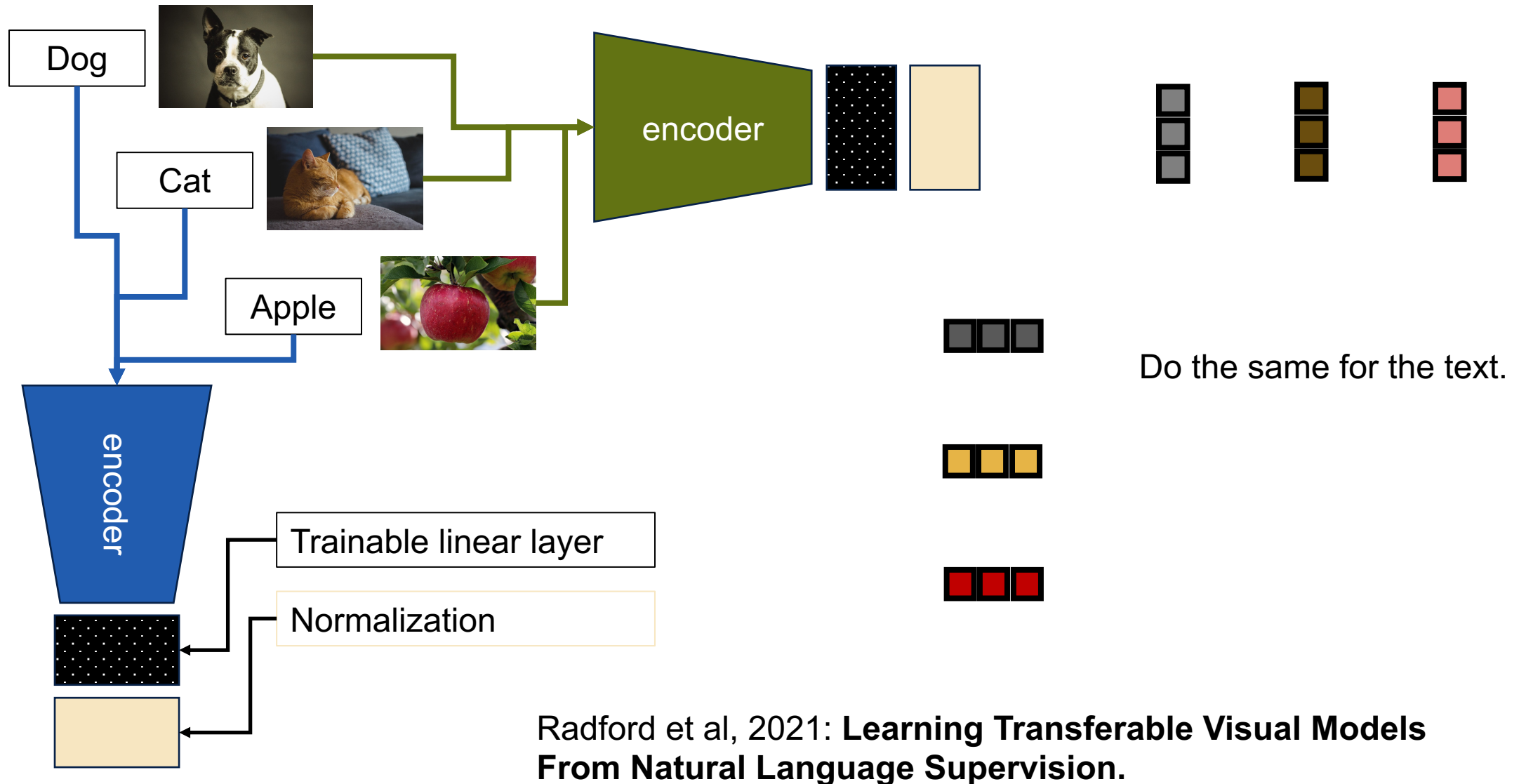
CLIP



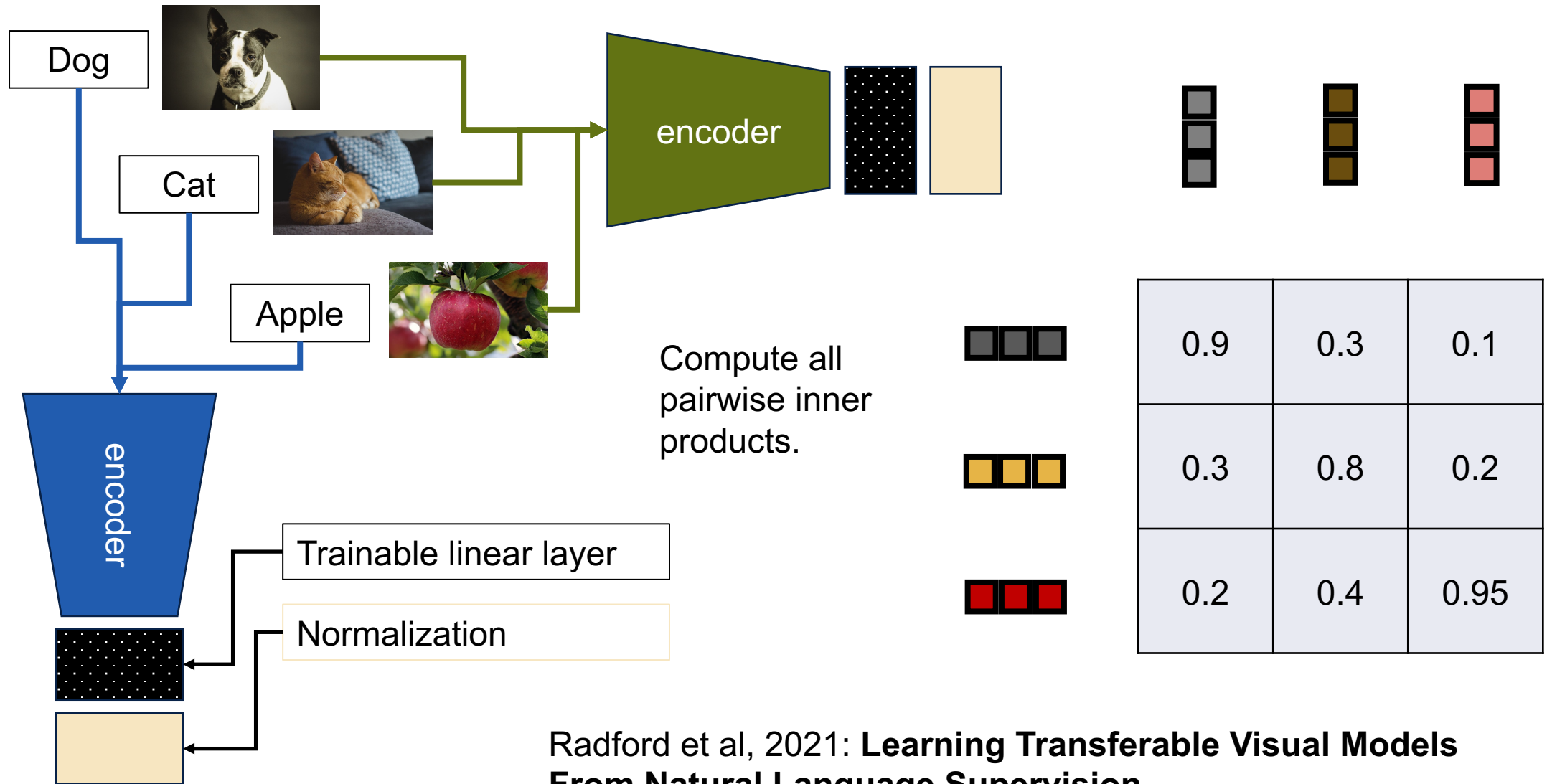
Compute a representation for each image.

Radford et al, 2021: **Learning Transferable Visual Models From Natural Language Supervision.**

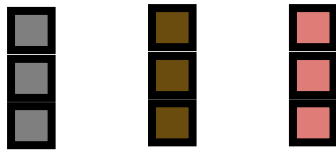
CLIP



CLIP



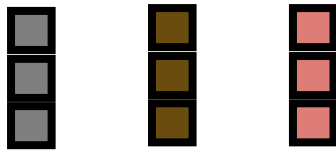
CLIP



0.9	0.3	0.1
0.3	0.8	0.2
0.2	0.4	0.95

$\times T$

CLIP



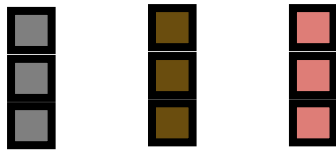
0.9	0.3	0.1
0.3	0.8	0.2
0.2	0.4	0.95

$\times T$

1	0	0
0	1	0
0	0	1

The resulting matrix should resemble the identity matrix.

CLIP

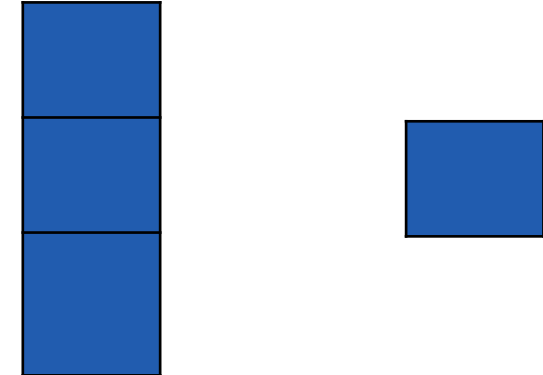


0.9	0.3	0.1
0.3	0.8	0.2
0.2	0.4	0.95

$\times T$

1	0	0
0	1	0
0	0	1

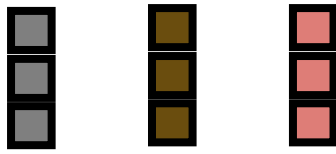
Row-wise losses



So we compute the cross entropy loss between the logits induced by each row and its corresponding row in the identity matrix.

Then we add the all these losses

CLIP



0.9	0.3	0.1
0.3	0.8	0.2
0.2	0.4	0.95

$\times T$

1	0	0
0	1	0
0	0	1

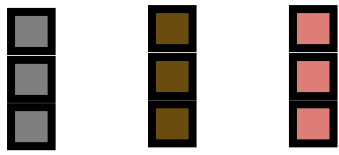


1	0	0
0	1	0
0	0	1

We do the same for the columns



CLIP



0.9	0.3	0.1
0.3	0.8	0.2
0.2	0.4	0.95

$\times T$

1	0	0
0	1	0
0	0	1

Row-wise losses



1	0	0
0	1	0
0	0	1

We then add the two losses and this is the target loss to minimize.

Column-wise losses

